

Indian Institute Of Technology Gandhinagar



Assignment Report

Student Name: Bilavath Prem Singh

Roll Number: 25210098

Branch: Integrated Circuit Design And Technology (ICDT)

Department: Electrical Engineering

Academic Year: 2025-2026

Submitted to: Madhav Pathak

Assignment No: 7

Date: march 24, 2026

Task 1: Designing a Low-Pass Filter using Ideal Op-Amp

1. (Refer to Fig. 1) Design a second-order unity-gain active low-pass filter with a cutoff frequency of 50 kHz, using the Sallen-Key topology.

Plot and compare the frequency response of the Butterworth, Bessel, and Chebyshev filters. Plot the time-domain response in the passband and stopband.

In this task, the design must be implemented using an ideal Op-Amp.

Also, report the roll-off (in dB/decade) for all three filters.

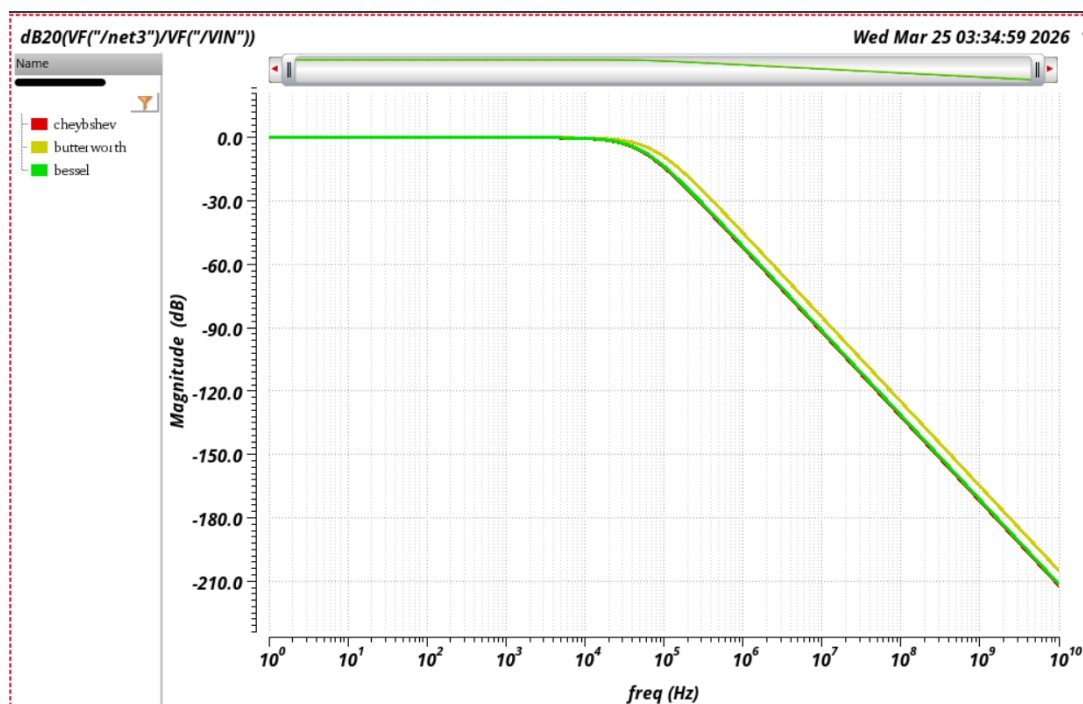


Figure 1: Ideal case comparison of three filters

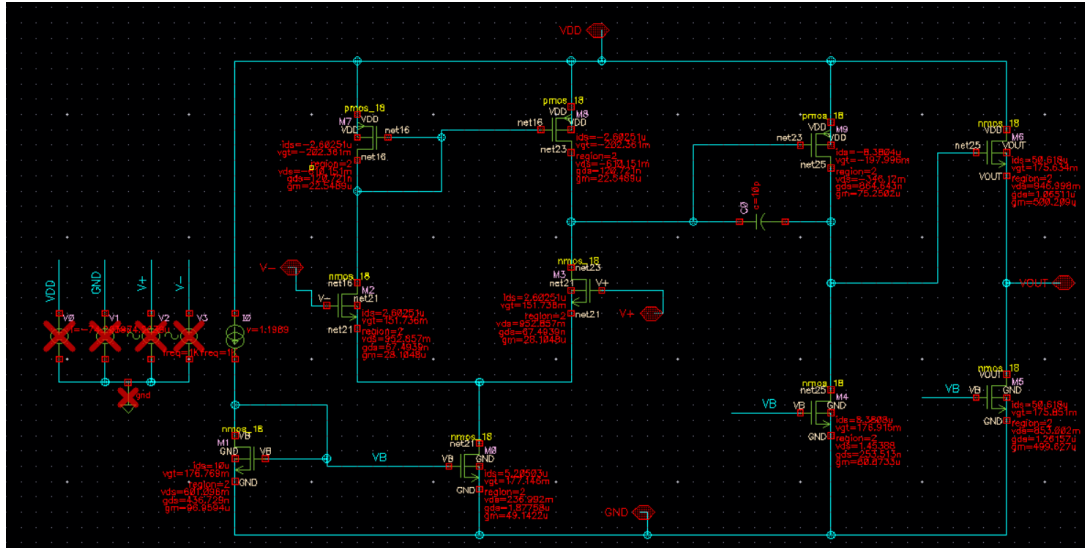


Figure 2: Opamp used in desin for sallen key filter

Table 1: Op-Amp Transistor Parameters

Transistor	V_{GT} (V)	g_m (S)	W (μm)
M1	176.75 mV	95.95 μS	2.165
M2	151.73 mV	28.10 μS	0.79
M3	151.73 mV	28.10 μS	0.79
M4	176.91 mV	80.07 μS	1.8
M5	176.91 mV	499.83 μS	10.5
M6	242.36 mV	500.21 μS	7.5
M7	202.36 mV	22.55 μS	8
M8	202.36 mV	22.55 μS	10.5
M9	197.99 mV	75.25 mS	8.2

Task 2: Designing a Low-Pass Filter using Designed Op-Amp [5 pts]

- (Refer to Fig. 1) Design a second-order unity-gain active low-pass filter with a cutoff frequency of 50 kHz, using the Sallen-Key topology.

Plot and compare the frequency response of the Butterworth, Bessel, and Chebyshev filters. Plot the time-domain response in the passband and stopband.

In this task, the design must be implemented using the designed Op-Amp.

Also, report the roll-off (in dB/decade) for all three filters.

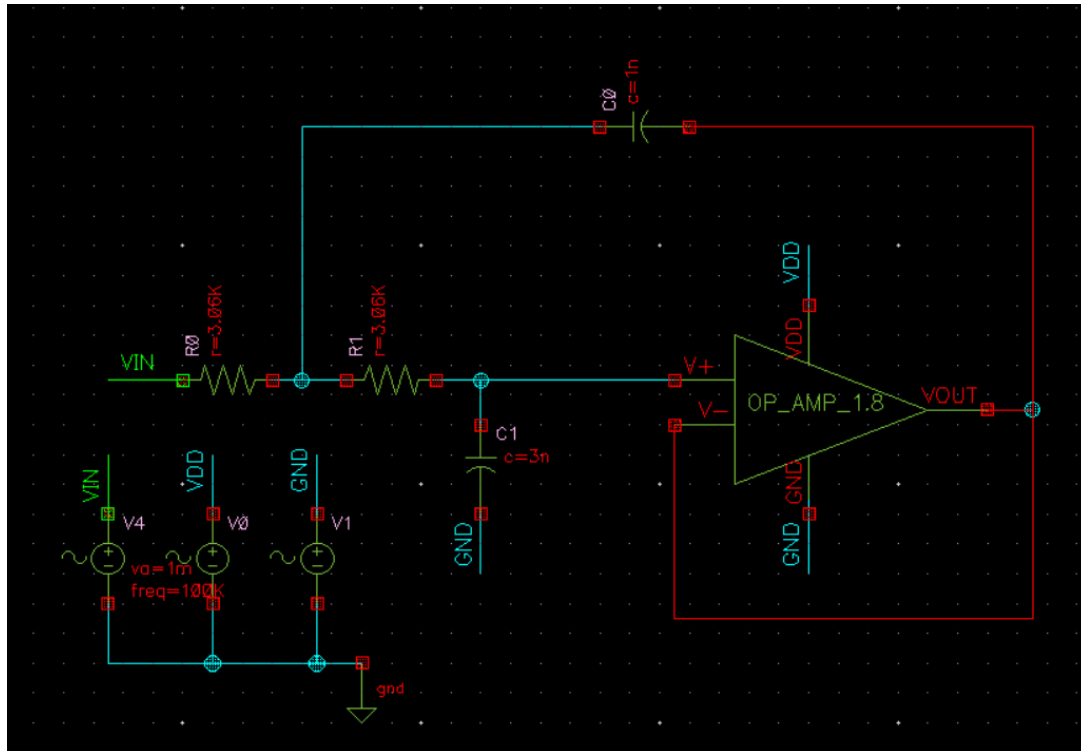


Figure 3: Circuit for sallen key filter topology

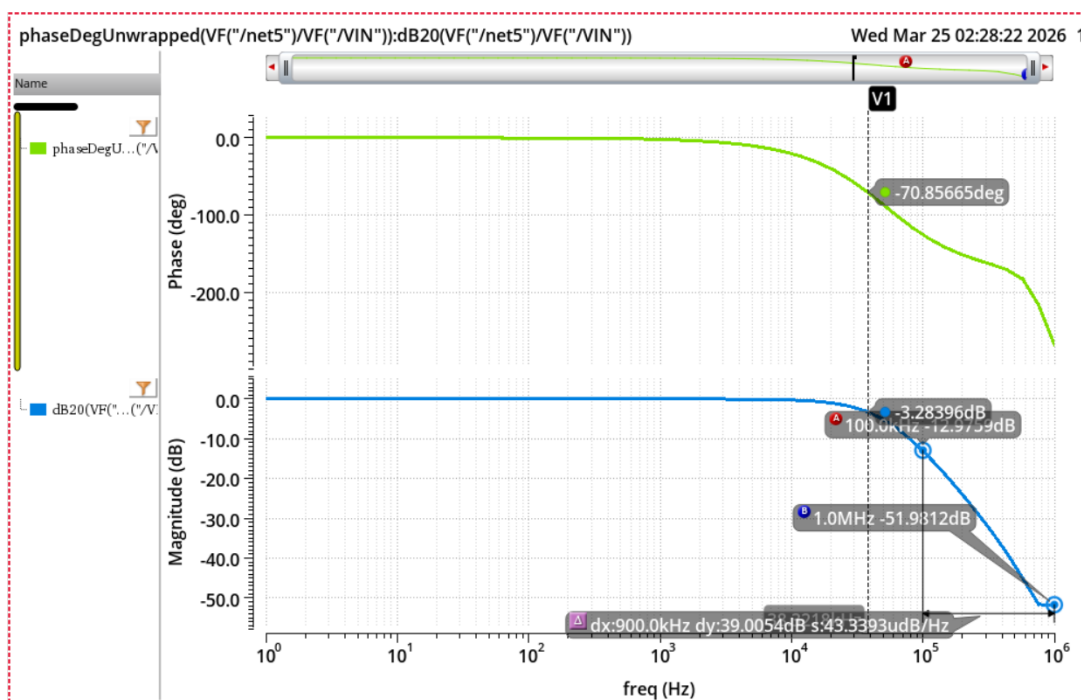


Figure 4: Butterworth filter bode plot

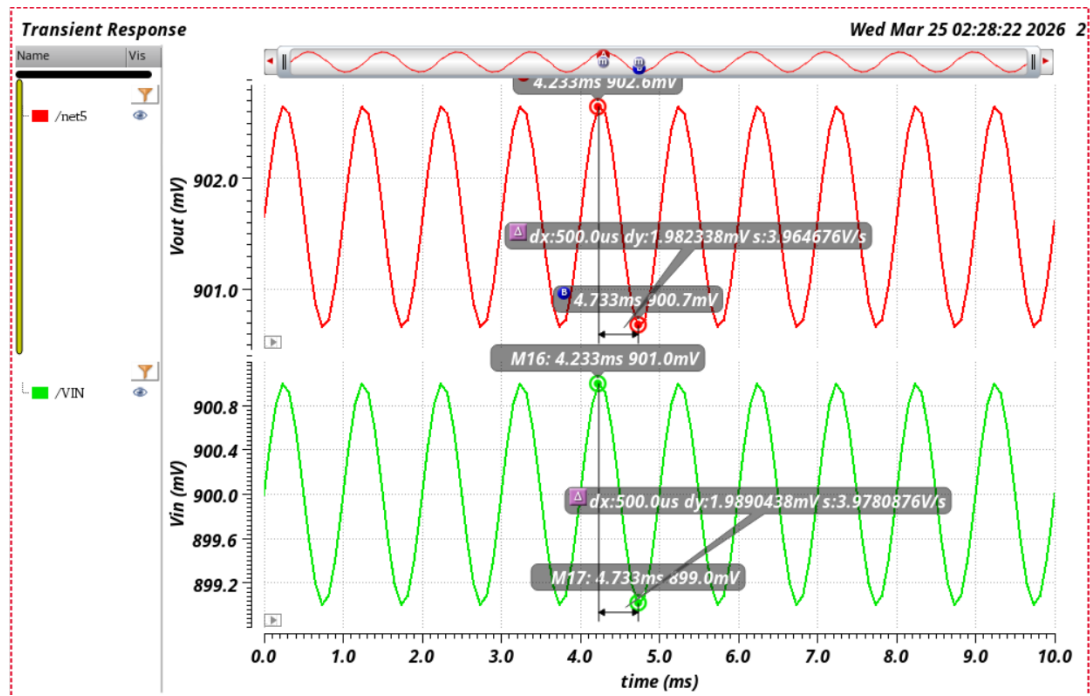


Figure 5: Butterworth pass band

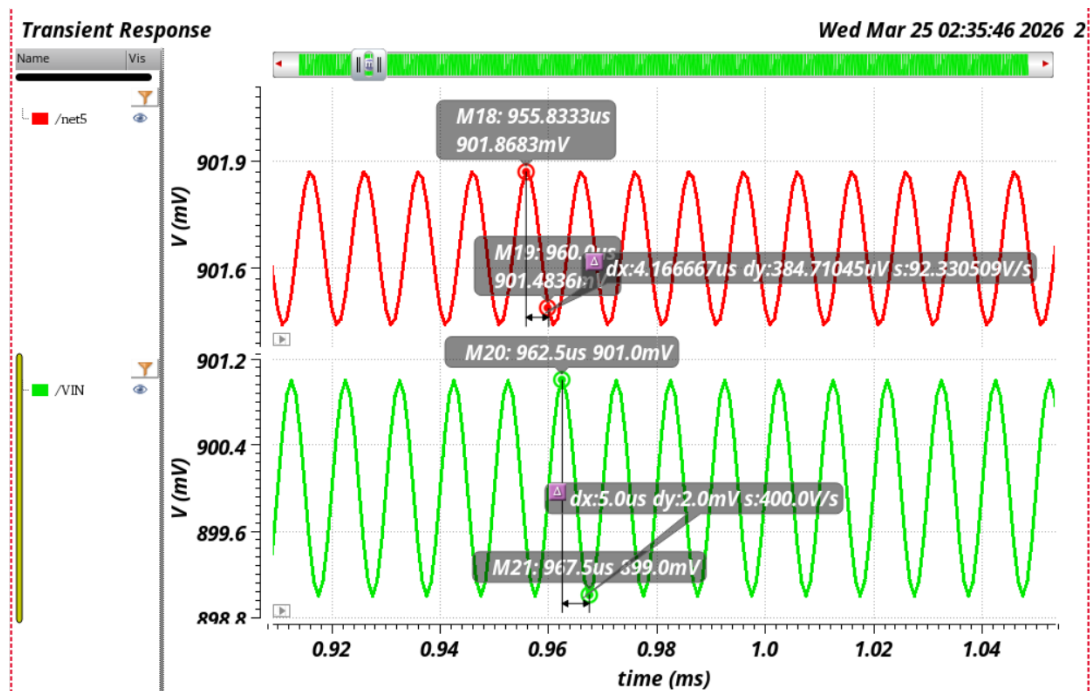


Figure 6: Butterworth stop band

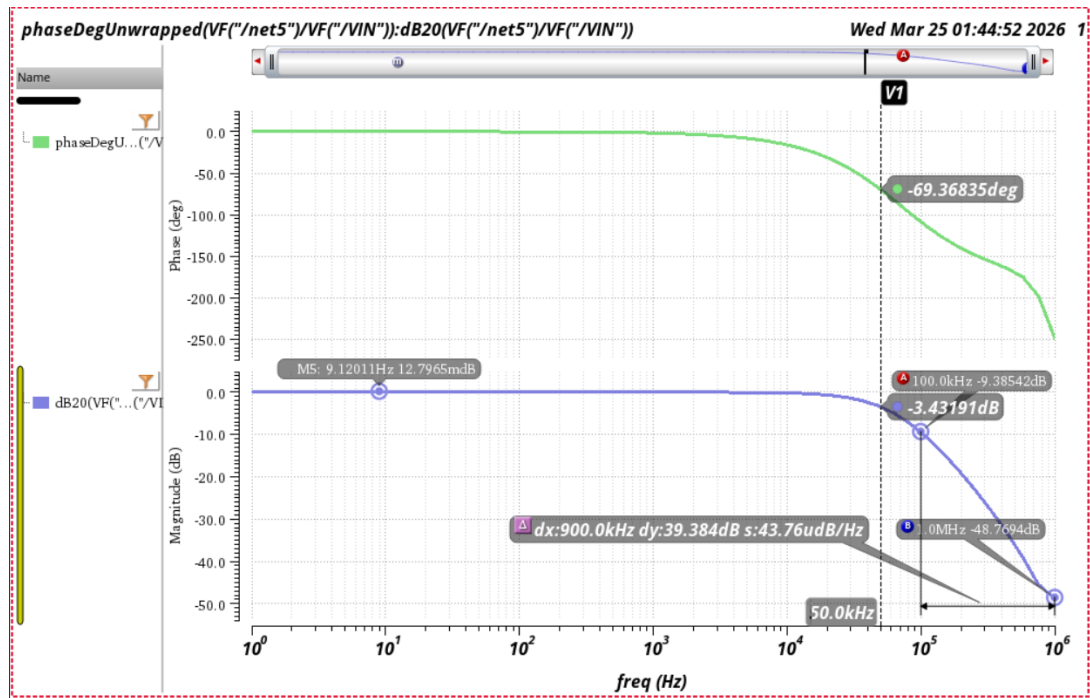


Figure 7: Bessel magnitude plot

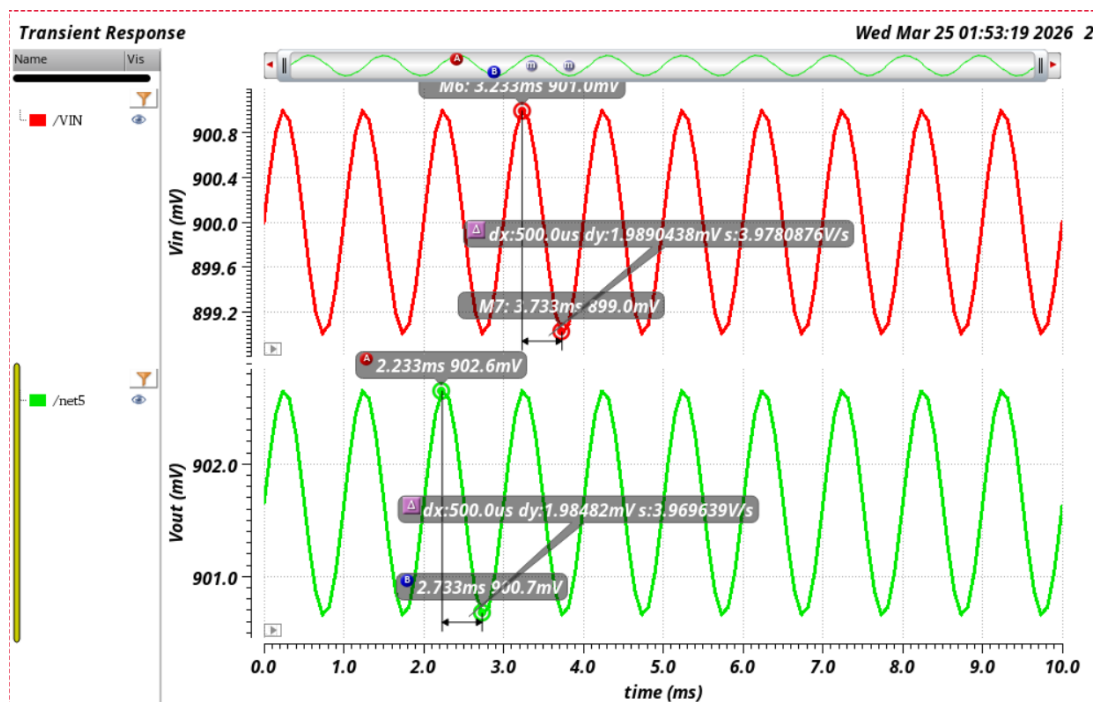


Figure 8: Bessel Pass band

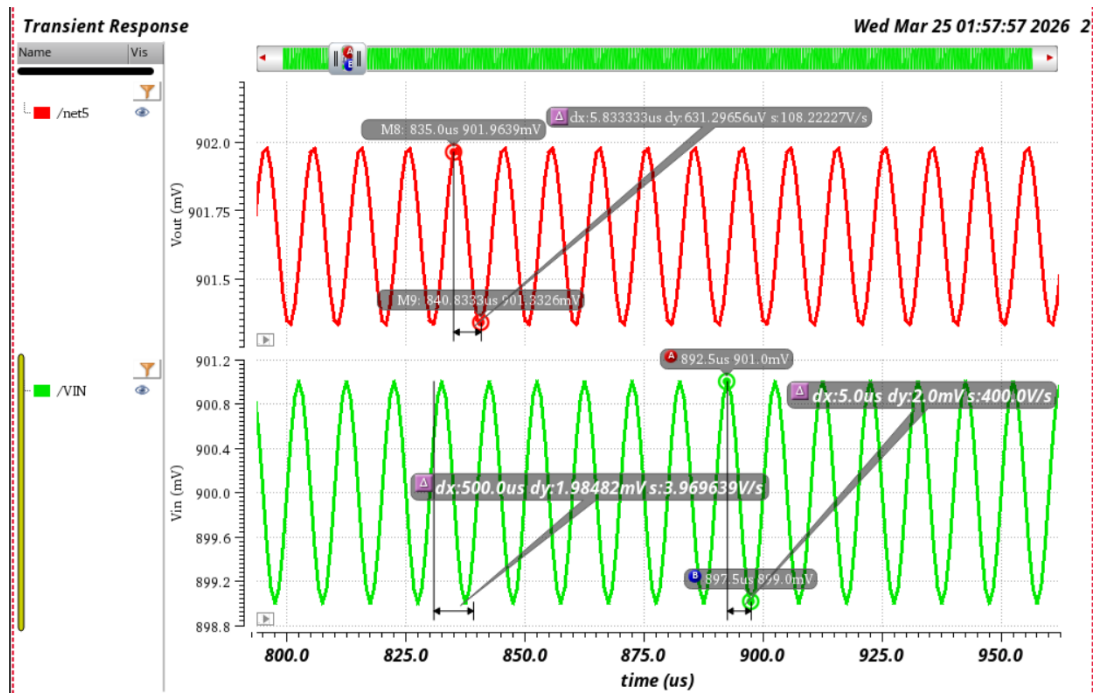


Figure 9: Bessel stop band

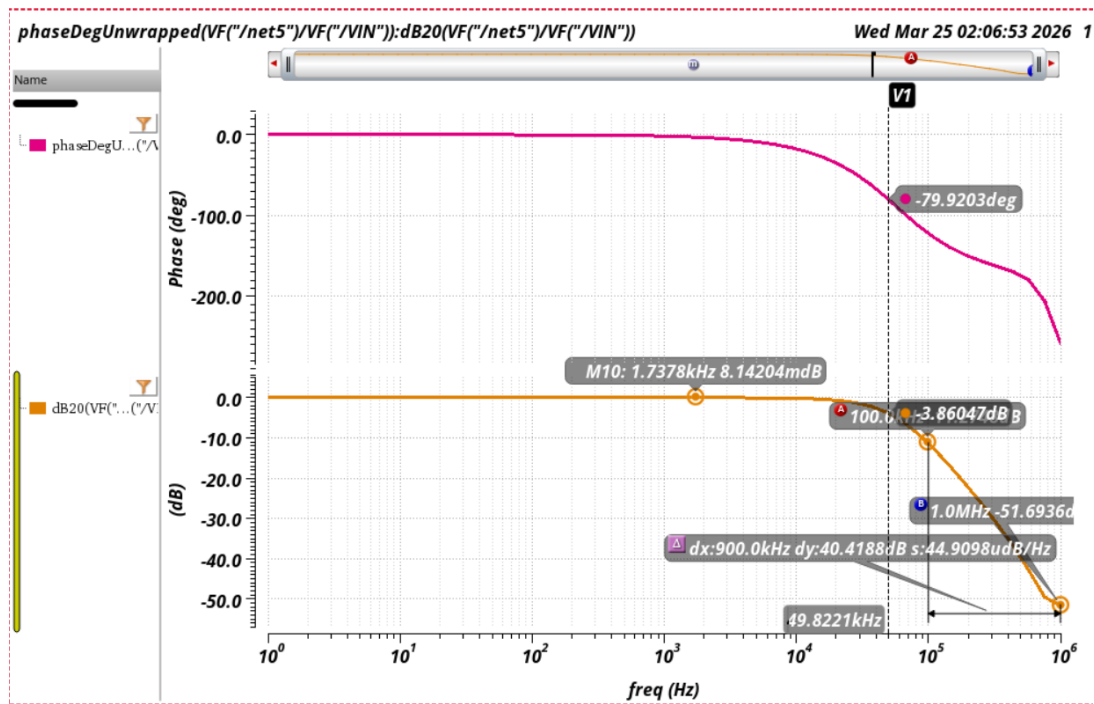


Figure 10: Chebyshev magnitude plot

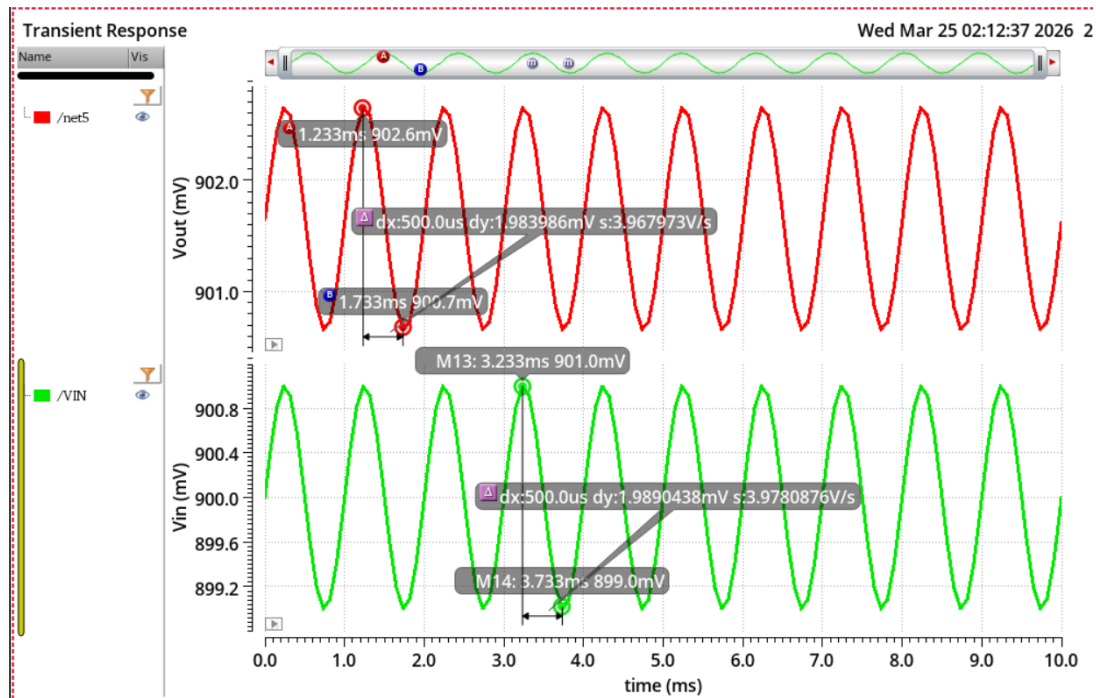


Figure 11: Chebyshev pass band

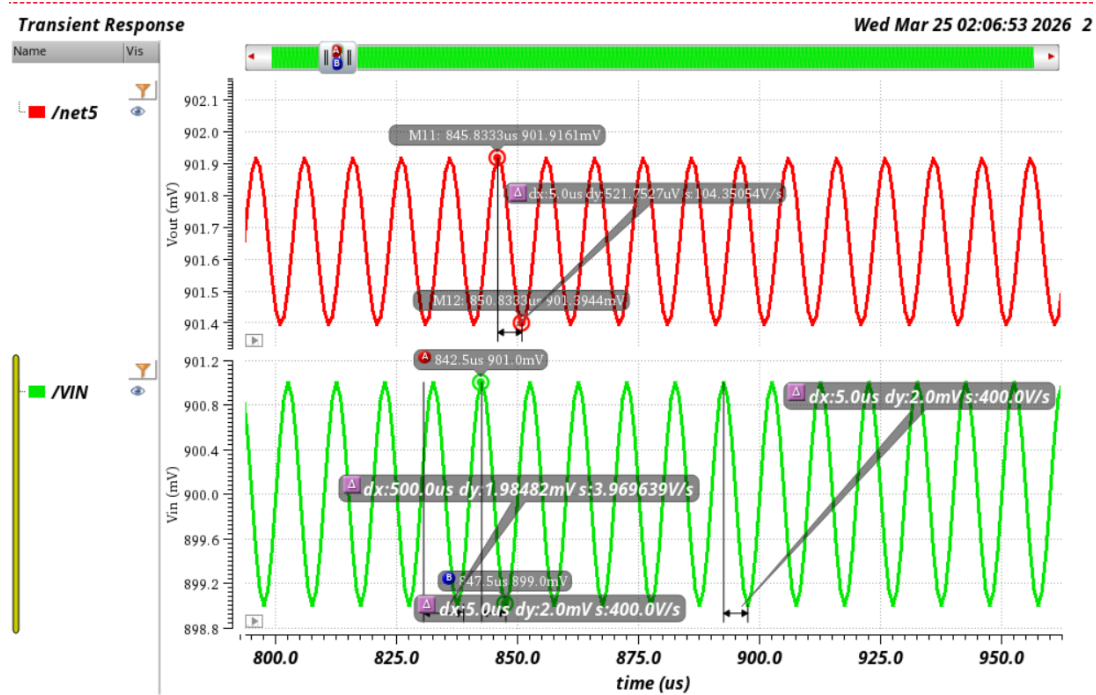


Figure 12: Chebyshev stop band

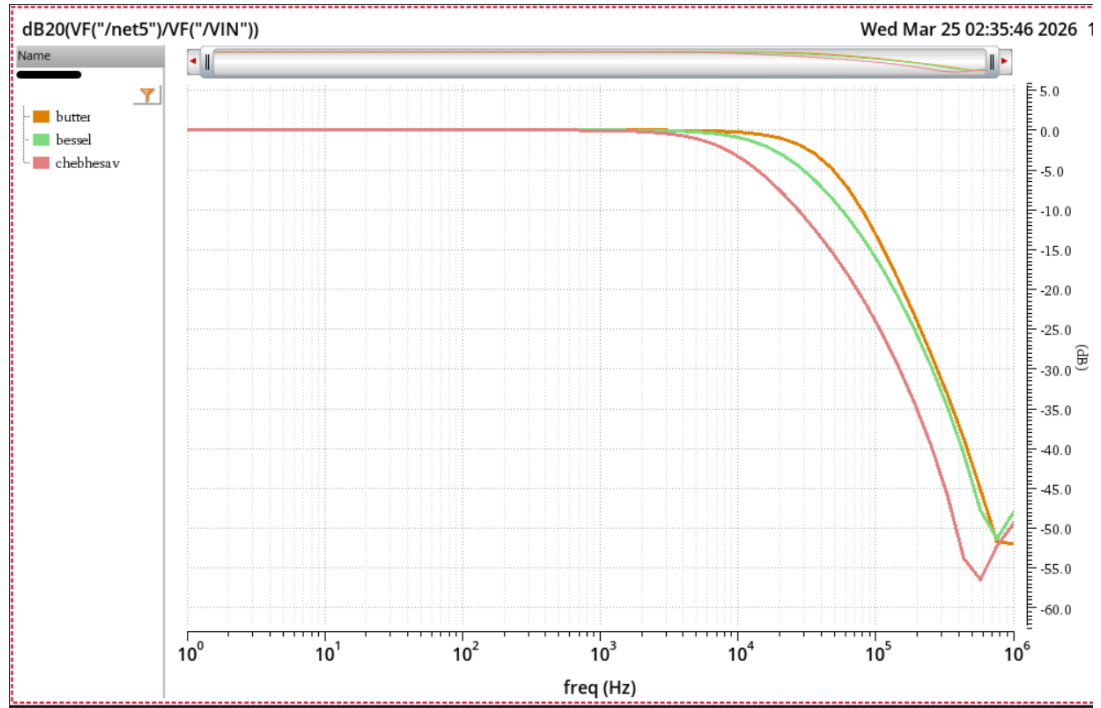


Figure 13: Comparison of three filters

Table 2: Cutoff Frequency and Roll-off Comparison from Bode Plots

Filter	Cutoff Frequency (f_c)	Gain at f_c (dB)	Roll-off (dB/dec)
Butterworth	50.0 kHz	-3.28 dB	-39.00
Bessel	50.0 kHz	-3.43 dB	-39.38
Chebyshev	49.82 kHz	-3.86 dB	-40.42

Filter Design Calculations

Given:

$$f_c = 50 \text{ kHz}$$

$$C_1 = C_2 = C = 1 \text{ nF} = 1 \times 10^{-9} \text{ F}$$

$$R_1 = R_2 = R$$

The design equation is:

$$FSF \cdot f_c = \frac{1}{2\pi RC}$$

Rearranging:

$$R = \frac{1}{2\pi \cdot C \cdot FSF \cdot f_c}$$

1. Butterworth Filter (FSF = 1)

$$R = \frac{1}{2\pi \cdot (1 \times 10^{-9}) \cdot (1) \cdot (50 \times 10^3)} \\ = 3.18 \text{ k}\Omega$$

2. Bessel Filter (FSF = 1.4524)

$$R = \frac{1}{2\pi \cdot (1 \times 10^{-9}) \cdot (1.4524) \cdot (50 \times 10^3)} \\ = 2.19 \text{ k}\Omega$$

3. Chebyshev Filter (FSF = 1.05)

$$R = \frac{1}{2\pi \cdot (1 \times 10^{-9}) \cdot (1.05) \cdot (50 \times 10^3)} \\ = 3.03 \text{ k}\Omega$$

Final Results

Filter Type	FSF	$R_1 = R_2$
Butterworth	1	3.18 k Ω
Bessel	1.4524	2.19 k Ω
Chebyshev	1.05	3.03 k Ω